

EMCH 260

SOLID MECHANICS

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Combined Loadings

8

Chapter 8.1

- ✓ Determine stresses developed in a member's cross section when axial load, torsion, bending and shear occur simultaneously

STATE OF STRESS CAUSED BY COMBINED LOADINGS

- In previous chapters, we developed methods for determining the stress distributions in a member subjected to internal axial forces, shear forces, bending moments, or torsional moments.
- Often, the x -section of a member is subjected to several of these type of loadings *simultaneously*
- We can use the method of **superposition** to determine the *resultant* stress distribution caused by the loads

STATE OF STRESS CAUSED BY COMBINED LOADINGS, CONT'D

Application of the method of superposition

1. The stress distribution due to each loading is determined *separately*
2. These distributions are *superimposed* to determine the total resultant stress distribution

Conditions to satisfy

- A linear relationship exists between the stress and the loads
- Geometry of the member should not undergo significant change when the loads are applied

CONDITIONS TO SATISFY, CONT'D

- This is necessary to ensure that the stress produced by one load is not related to the stress produced by any other load
 - *For example, this tall chimney is subjected to the combined internal loading caused by the wind (lateral force) and the chimney's weight (vertical force)*



PROCEDURE FOR ANALYSIS

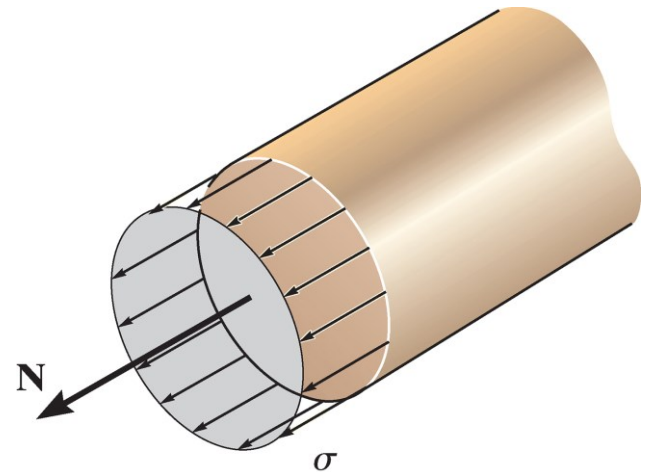
Step 1: Internal Loading

- Section the member perpendicular to its axis at the point where the stress is to be determined
- Obtain the resultant internal normal and shear force components and the bending and torsional moment components
- Force components should act through the **centroid** of the x -section, and moment components should be computed about **centroidal axes**, which represent the principal axes of inertia for x -section

Step 2: Stress Components

Normal stress

- Internal normal force is developed by a uniform normal-stress distribution by $\sigma = N/A$



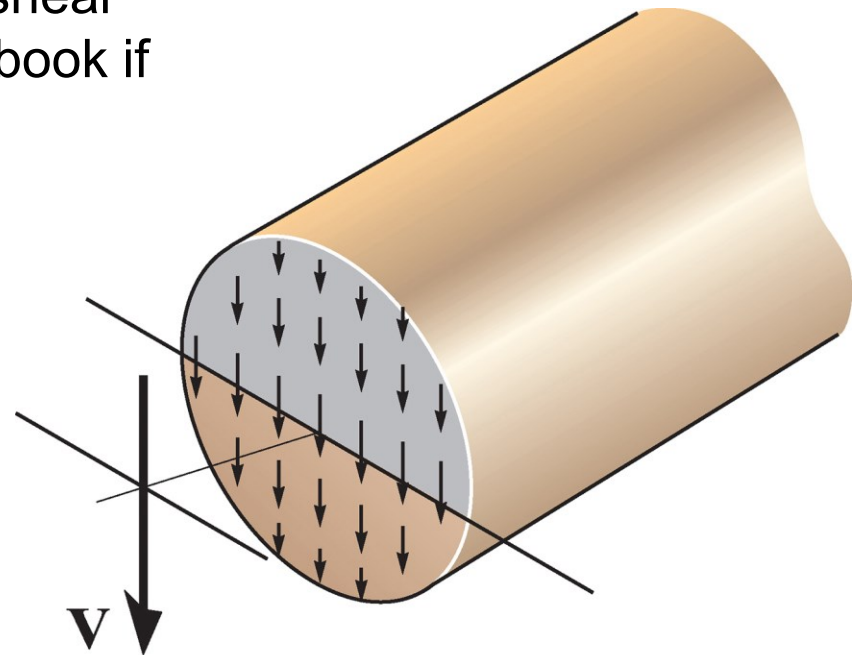
$$\sigma = \frac{N}{A}$$

PROCEDURE FOR ANALYSIS, CONT'D

Step 2: Stress Components, Cont'd

Shear force

- Internal shear force in member subjected to bending is developed from shear-stress distribution determined from the shear formula (refer to page 370 in textbook if necessary), $\tau = VQ/It$.



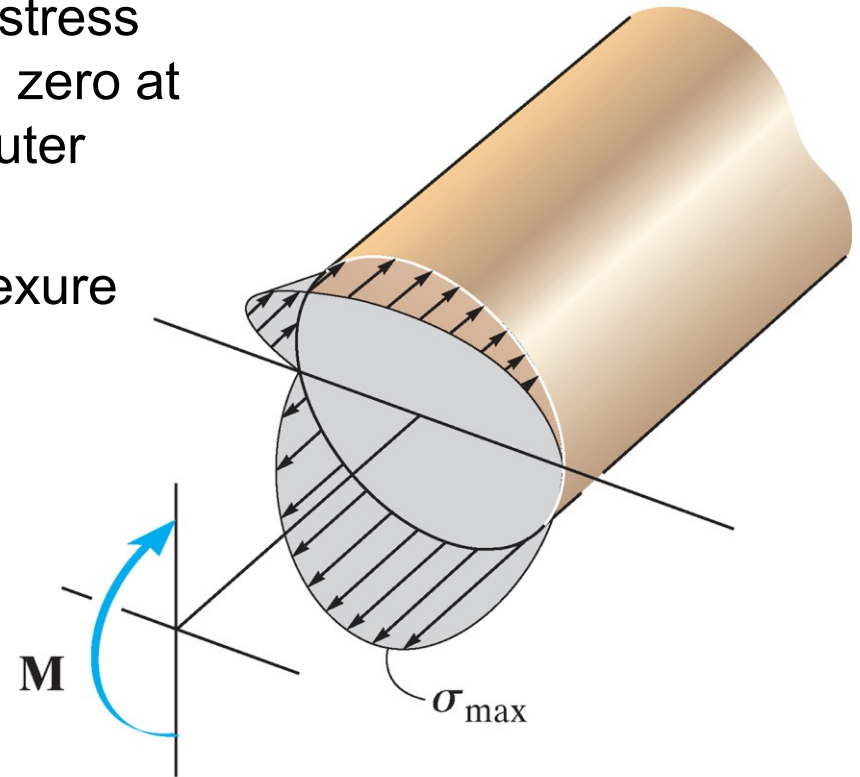
$$\tau = \frac{VQ}{It}$$

PROCEDURE FOR ANALYSIS, CONT'D

Step 2: Stress Components, Cont'd

Bending moment

- For straight members, the internal bending moment is developed by a normal-stress distribution that varies linearly from zero at the neutral axis to a maximum at outer boundary of the member.
- Stress distribution obtained from flexure formula, $\sigma = -My/I$.



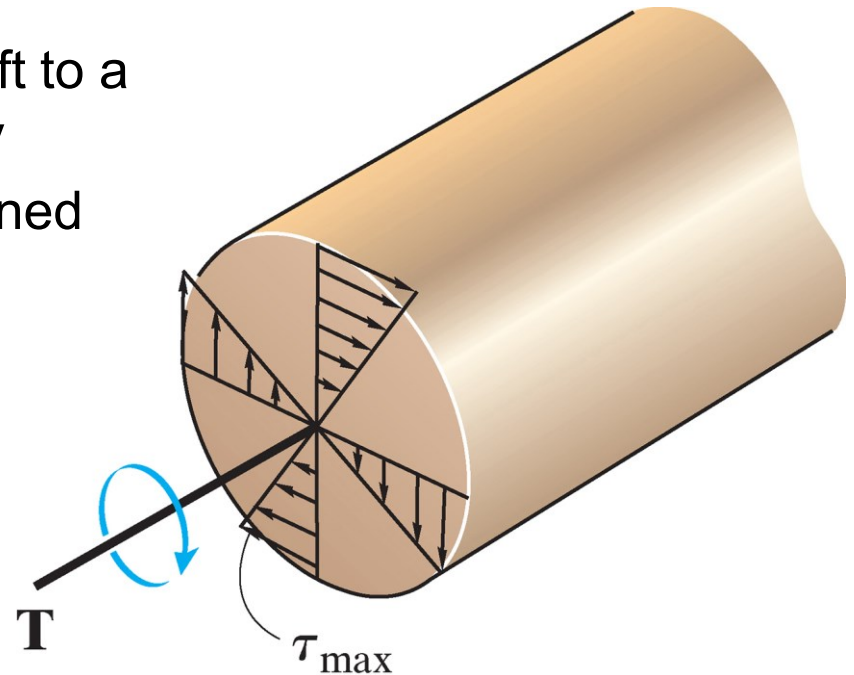
$$\sigma = -\frac{My}{I}$$

PROCEDURE FOR ANALYSIS, CONT'D

Step 2: Stress Components, Cont'd

Torsional moment

- For circular shafts and tubes, internal torsional moment is developed by a shear-stress distribution that varies linearly from the central axis of shaft to a maximum at shaft's outer boundary
- Shear-stress distribution is determined from the torsional formula, $\tau = T\rho/J$.



$$\tau = \frac{T\rho}{J}$$

PROCEDURE FOR ANALYSIS, CONT'D

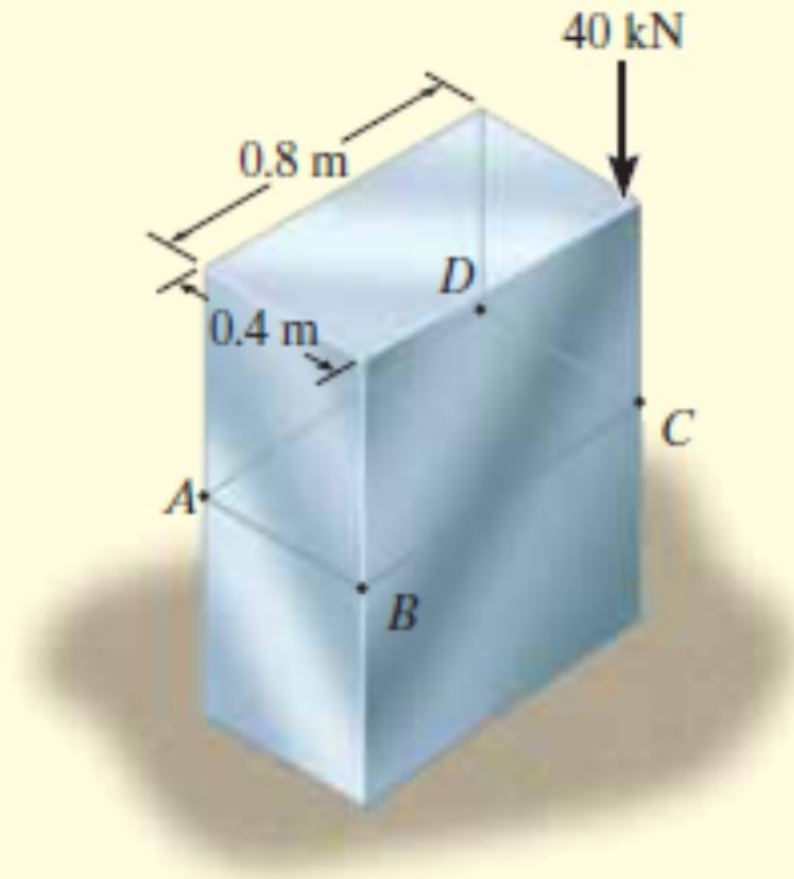
Step 3: Superposition

- Once normal and shear stress components for each loading have been calculated, use the *principle of superposition* and determine the resultant normal and shear stress components
- Represent the results on an element of material located at the point, or show the results as a distribution of stress acting over the member's x -sectional area

NEVER ADD DIFFERENT TYPES OF STRESS TOGETHER!!!

IN-CLASS EXAMPLE 1

The rectangular block of negligible weight is subjected to a vertical force of 40 kN, which is applied to its corner. Determine the state of stress at each point A , B , C , and D , and the largest normal stress acting on the section through $ABCD$.



IN-CLASS EXAMPLE 1, Cont'd

Solutions

Step 1: Internal Loading

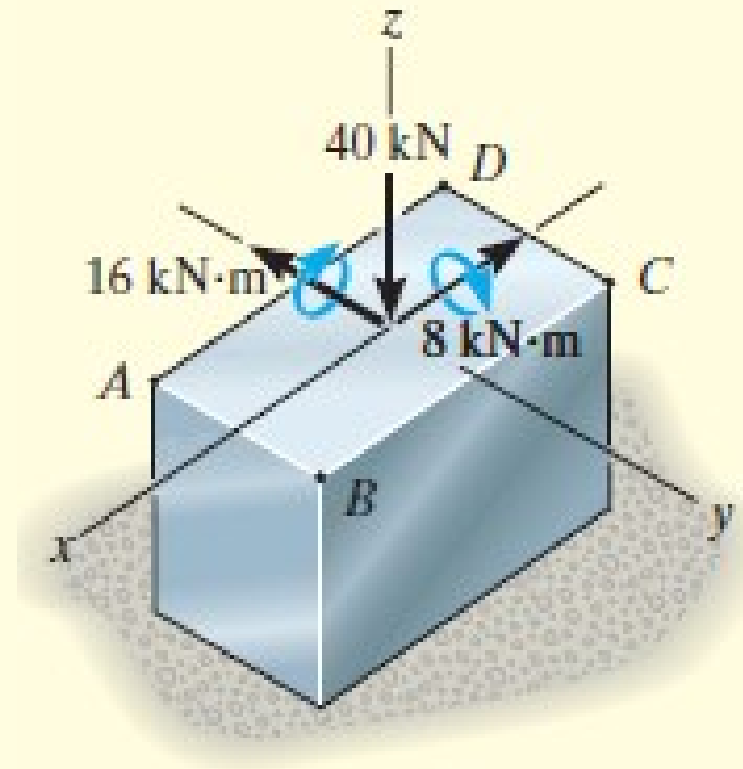
Section the member along the $ABCD$ plane, where the stress is to be determined. The xyz coordinate system is established with origin at the centroid point of $ABCD$.

The resultant internal normal and shear force components and the bending and torsional moment components can be obtained as:

- Normal force in z -direction: $P = 40 \text{ kN}$
- Bending moment components:

$$M_x = 40000 \times \frac{0.4}{2} = 8 \text{ kN} \cdot \text{m}$$

$$M_y = 40000 \times \frac{0.8}{2} = 16 \text{ kN} \cdot \text{m}$$



IN-CLASS EXAMPLE 1, Cont'd

Solutions, Cont'd

Step 2: Stress Components

- For normal force in z-direction the uniform normal-stress is

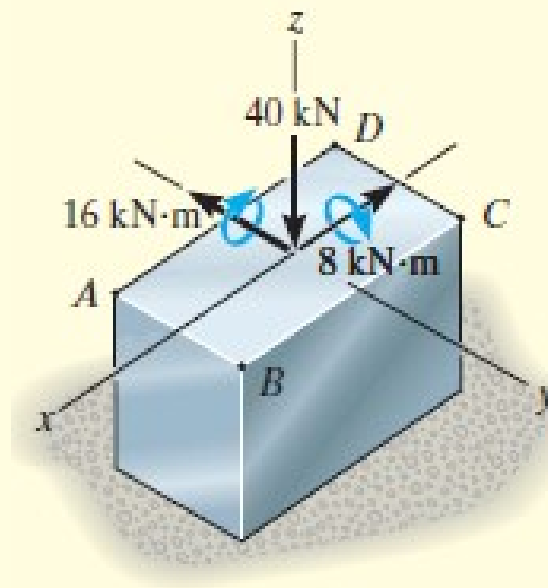
$$\sigma = \frac{P}{A} = \frac{40}{(0.8)(0.4)} = 125 \text{ kPa}$$

- For moment in x-direction, the maximum stress is

$$\sigma_{\max} = \frac{M_x c_x}{I_x} = \frac{8(0.2)}{\left[\frac{1}{12}(0.8)(0.4)^3\right]} = 375 \text{ kPa}$$

- For moment in y-direction, the maximum stress is

$$\sigma_{\max} = \frac{M_y c_y}{I_y} = \frac{16(0.4)}{\left[\frac{1}{12}(0.4)(0.8)^3\right]} = 375 \text{ kPa}$$

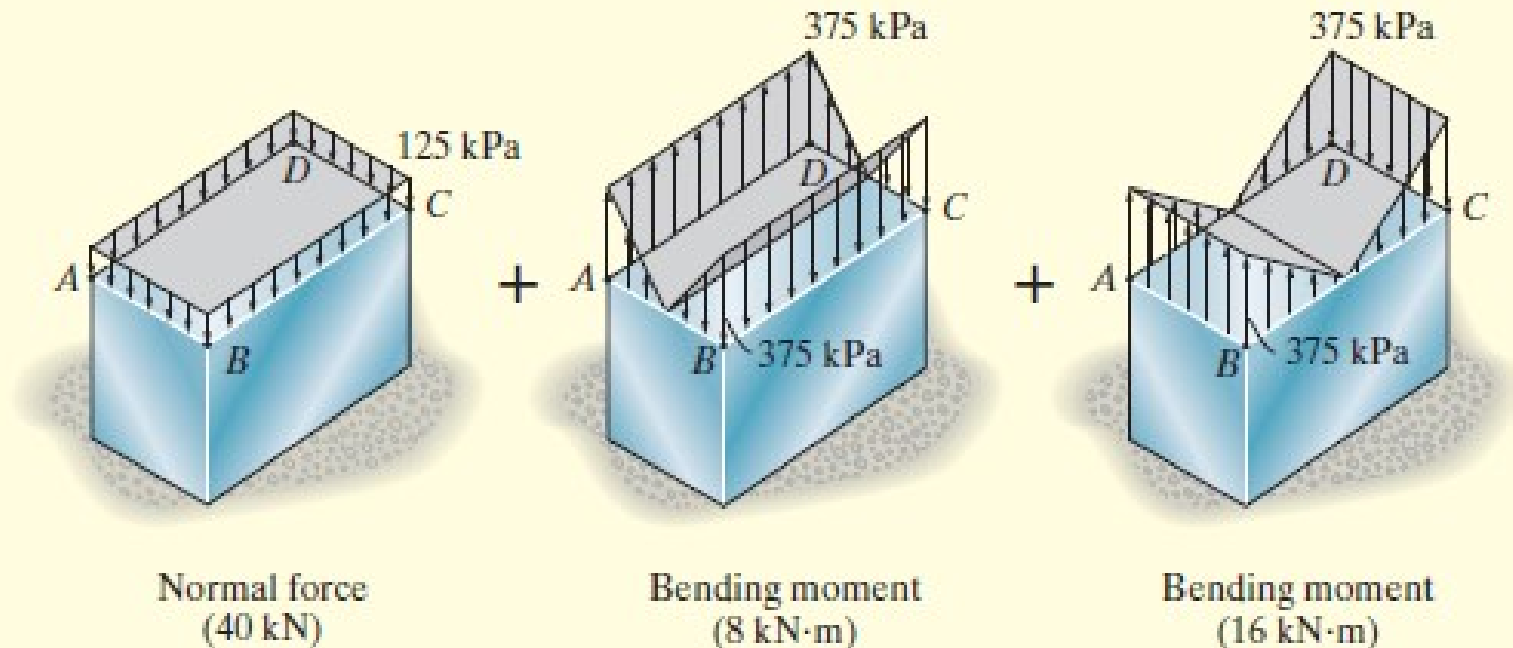


IN-CLASS EXAMPLE 1, Cont'd

Solutions, Cont'd

Step 3: Superposition

The total resultant normal stress is the summation of all the components induced by normal force in z-direction, bending moment along x- and y-direction.



The normal stress at point C is the largest, since each loading creates a **compressive** stress there, i.e.

$$\sigma_c = -125 - 375 - 375 = -875 \text{ kPa} \quad (\text{Ans})$$

RELEVANT PRACTICE

- **Assignment 3 on Blackboard, problem #1-2 (go to Blackboard under the “Assignments” folder, **Assignment 3 due: 8:30 AM, Friday, Nov. 5th 2021**)**
- **Optional Extra Exercise 3 on Blackboard, problem #1 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 3 due: 8:30 AM, Friday, Nov. 5th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 413 – 428**
 - **In particular, Examples 8.2, 8.3, and 8.7 (pp. 422, 423, and 428)**